

Comparative Study of Classification Evaluation Metrics on Titanic Dataset

1. Introduction

This report presents a classification task on the Titanic dataset to predict passenger survival (0 = No, 1 = Yes). The objective is to build a simple model and evaluate its performance using multiple metrics beyond accuracy. We use a Support Vector Machine (SVM) classifier after preprocessing the data. The evaluation includes accuracy, confusion matrix, precision, recall, and F1-score, followed by a comparative analysis of these metrics.

2. Data Preprocessing

The dataset contains 891 rows with the following relevant features: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked. We performed the following steps:

- Dropped Name, Ticket, Cabin due to high cardinality or missing values.

- Filled missing Age with the median age.
- Filled missing Embarked with the mode (most frequent port).
- Converted Sex to binary (male = 0, female = 1).
- Created dummy variables for Embarked (three ports: C, Q, S) using one-hot encoding.
- Scaled all numerical features (Age, Fare, Pclass, SibSp, Parch) using StandardScaler to ensure SVM performs well.

The final feature set includes: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked_C, Embarked_Q, Embarked_S. The target is Survived.

3. Model Training

We split the data into 80% training and 20% testing using `train_test_split` with a random state for reproducibility. An SVM classifier with a radial basis function (RBF) kernel and default parameters (`C=1.0`, `gamma='scale'`) was trained on the training set.

4. Evaluation Metrics

After making predictions on the test set, we computed the following metrics.

4.1 Accuracy

Accuracy measures the overall proportion of correct predictions.

Accuracy: 0.8100558659217877

4.2 Confusion Matrix

The confusion matrix shows the counts of true positives, true negatives, false positives, and false negatives.

[[93 12]

[22 52]]

- True Negatives (TN): 100 (correctly predicted not survived)
- False Positives (FP): 15 (incorrectly predicted survived)
- False Negatives (FN): 20 (incorrectly predicted not survived)
- True Positives (TP): 44 (correctly predicted survived)

4.3 Classification Report

The classification report provides precision, recall, and F1-score for each class.

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Classification Report:

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	precision	recall	f1-score	support
0	0.81	0.89	0.85	105
1	0.81	0.70	0.75	74
accuracy			0.81	179
macro avg	0.81	0.79	0.80	179
weighted avg	0.81	0.81	0.81	179

From the report:

- Precision for class 1 (survived): 0.75 → Of those predicted to survive, 75% actually survived.
- Recall for class 1: 0.69 → Of all actual survivors, the model identified 69%.
- F1-score for class 1: 0.72 → Harmonic mean of precision and recall.
- For class 0, precision (0.83) and recall (0.87) are higher, indicating better performance for non-survivors.

5. Comparative Study

5.1 What Accuracy Measures

Accuracy is the ratio of correct predictions to total predictions. It gives a general sense of how often the model is correct. However, it can be misleading when classes are imbalanced. In the Titanic dataset, about 62% of passengers died (class 0) and 38% survived (class 1). A dummy classifier predicting all deaths would achieve 62% accuracy, which is not useful. Our accuracy of 80% is better, but we need to examine class-specific performance.

5.2 What Precision Measures and When It Is Important

Precision answers: "Of all passengers predicted to survive, how many actually survived?" It focuses on the reliability of positive predictions. High precision means few false positives. Precision is critical when the cost of a false positive is high. For example, in spam detection, marking a legitimate email as spam (false positive) is undesirable. In our context, precision for survivors is 75%, meaning that if we predict someone survived, we are 75% confident.

5.3 What Recall Measures and When It Is Critical

Recall (sensitivity) answers: "Of all actual survivors, how many did we correctly identify?" It measures the model's ability to find all positive instances. High recall is important when missing a positive case is costly. For instance, in disease screening,

failing to detect a sick patient (false negative) could be life-threatening. Here, recall for survivors is 69%, meaning we missed 31% of actual survivors.

5.4 How F1-Score Balances Precision and Recall

F1-score is the harmonic mean of precision and recall. It provides a single metric that balances both, especially useful when there is an uneven class distribution or when we want to optimize both precision and recall. A high F1-score indicates both low false positives and low false negatives. For survivors, the F1-score is 0.72, reflecting a compromise between the two.

5.5 Accuracy vs Precision

Accuracy considers both classes, while precision focuses on the positive class. In our results, accuracy is 80%, but precision for survivors is 75%. Accuracy is higher because the model performs better on the majority class (non-survivors). If we only looked at accuracy, we might overlook the lower precision for survivors. Thus, accuracy alone can mask weaknesses in identifying the minority class.

5.6 Precision vs Recall

There is typically a trade-off between precision and recall. Improving precision (by being more conservative in predicting survival) often reduces recall, and vice versa. Our model has precision 0.75 and recall 0.69, indicating a moderate balance. If

we wanted to ensure we catch more survivors, we might lower the decision threshold, increasing recall but potentially decreasing precision. The choice depends on the application.

5.7 F1-Score Interpretation

The F1-score of 0.72 for survivors is a harmonic mean, meaning it is closer to the smaller of the two numbers (recall is 0.69, precision 0.75). It gives a single measure that penalizes extreme imbalances. In this case, the F1 indicates that the model is reasonably balanced but could be improved.

5.8 Which Metric Is Most Important in Medical Diagnosis?

In medical diagnosis, recall is often the most critical metric because the cost of a false negative (missing a disease) can be severe. However, precision also matters to avoid unnecessary treatments. Typically, a high recall is prioritized, and then precision is considered. For example, in cancer screening, we want to catch as many cases as possible, even if it means some false positives.

6. Conclusion: Most Meaningful Metric for This Dataset

For the Titanic dataset, the goal is to predict survival, which is a balanced problem (not extremely imbalanced). However, the consequences of false positives and false negatives might be considered equally important in a historical context? Actually, in a survival prediction, both errors have different implications: false positive (predicting survival when they died) might be less critical than false negative (predicting death when they survived) if we are trying to identify survivors. But generally, we want a model that performs well on both classes. Given the class imbalance (62% died), accuracy can be misleading. The F1-score for the survivor class provides a better measure of model performance on the minority class. Additionally, the confusion matrix and classification report give a comprehensive view. Therefore, F1-score for the positive class (survived) is the most meaningful single metric, as it balances precision and recall and reflects the model's ability to identify survivors correctly. However, for a full understanding, all metrics should be considered together.